$$\frac{i \in \{1,2\}}{x_i : A \vdash (x_1 =_A x_2) :: x_{3-i} : A} \qquad \frac{g : T \to \bigoplus_{j=1}^{k} U_j \quad 1 \le j \le k}{\vec{x} : T \vdash (g^j(\vec{x}) = \vec{y}) :: \vec{y} : U_j} \qquad \frac{}{\vdash 1 ::} \qquad \frac{\Gamma \vdash p :: \Delta}{\Gamma, x : A \vdash p :: \Delta}$$